

Monitoring of buccal epithelial cells by alkaline comet assay (single cell gel electrophoresis technique) in cytogenetic evaluation of chlorhexidine.

During the last few years, there has been increasing interest in buccal epithelial cells for cytogenetic evaluation of different materials. In the present study, the use of these cells and peripheral lymphocytes for cytogenetic evaluation of chlorhexidine digluconate (CHX) with comet assay (single cell gel electrophoresis, or SCGE) is reported. This technique detects DNA strand breaks in individual cells in alkaline conditions. Thirteen volunteers were requested to rinse their mouths with 0.12% CHX solution for 18 days. Buccal epithelial cells and peripheral blood lymphocytes were obtained from all participants at baseline and the end of the experimental period. One hundred cells per subject were analysed for the DNA damage. A statistical increase was observed in the damaged buccal and blood cells after the CHX application. The mean grade of damage in buccal cells was statistically different from that in blood cells. Due to minimal absorption of chlorhexidine into the tissues and low concentrations of free chlorhexidine in the oral cavity, the DNA damage produced by chlorhexidine in lymphocytes was lower than in buccal epithelial cells. As chlorhexidine does not accumulate in the body, the frequencies of DNA damage could be transient. Detected DNA damage after CHX use might be the indication of an earlier effect, before DNA repair begins, and could be reversible.